3.1 Law of Sines

Name _____ Period ____ Date _____

Pre-Calculus 3.1 (30-minute plan) · Student sheet

I can…

- use the Law of Sines to solve oblique triangles (AAS or ASA).
- use the Law of Sines to solve oblique triangles (SSA), including the ambiguous case.
- find the area of an oblique triangle and solve real-life problems.

Part 1 · Notes: fill in the blanks

Law of Sines: _____ = _____ = _____

Use the Law of Sines for _____, _____, and _____.

In the SSA case, if sin B > 1 there is _____. If sin B < 1, check whether _____ also works.

Area of an oblique triangle = _____

Part 2 · Examples (work them with the class)

Example 1

Solve the triangle: $C = 102.4°$, $B = 28.7°$, $b = 27.4$ feet. Round to two decimal places.

Example 2

Find two triangles with $A = 20.5°$, $a = 12$, and $b = 31$. Round to two decimal places.

Example 3

Find the area of a triangular lot with sides of 52 meters and 90 meters and an included angle of 102°.

Part 3 · Partner practice

Partner practice

How many triangles have A = 30°, a = 6, and b = 10? Find B for each.